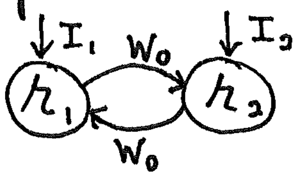
# Symmetric two neuron example:

Let's practice and build intuition with an example.




$$; \quad W = \begin{pmatrix} 0 & W_0 \\ W_0 & 0 \end{pmatrix}$$

The coupled equations are,

$$\tau \frac{dr_1}{dt} = -r_1 + W_0 r_2 + I_1$$

$$\tau \frac{dr_2}{dt} = -r_2 + W_0 r_1 + I_2$$

Find eigenvalues and eigenvectors of W to decouple & solve.

$$0 = \det(W - \lambda I) = \det\begin{pmatrix} -\lambda & W_0 \\ W_0 & -\lambda \end{pmatrix} = \lambda^2 - W_0^2$$

$$\lambda_\pm = \pm W_0$$

$$W \xi^{(\pm)} = \begin{pmatrix} 0 & W_0 \\ W_0 & 0 \end{pmatrix} \begin{pmatrix} \xi_1^{(\pm)} \\ \xi_2^{(\pm)} \end{pmatrix} = \begin{pmatrix} W_0 \xi_2^{(\pm)} \\ W_0 \xi_1^{(\pm)} \end{pmatrix} = \pm W_0 \begin{pmatrix} \xi_1^{(\pm)} \\ \xi_2^{(\pm)} \end{pmatrix}$$

$$\xi_2^{(\pm)} = \pm \xi_1^{(\pm)}, \quad \underbrace{\xi^{(+)} = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix}}_{\text{common mode}}, \quad \underbrace{\xi^{(-)} = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ -1 \end{pmatrix}}_{\text{difference mode}}$$

$$V = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}, \quad V^{-1} = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$$

$$\beta = \begin{pmatrix} \beta_+ \\ \beta_- \end{pmatrix} = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} \begin{pmatrix} I_1 \\ I_2 \end{pmatrix} = \frac{1}{\sqrt{2}} \begin{pmatrix} I_1 + I_o \\ I_1 - I_2 \end{pmatrix}$$

The two uncoupled equations become

$$\tau \frac{d\alpha_+}{dt} = -\alpha_+ (1 - W_0) + \frac{I_1 + I_2}{\sqrt{2}}$$

$$\tau \frac{d\alpha_-}{dt} = -(1 + W_0)\alpha_- + \frac{I_0 - I_2}{\sqrt{2}}$$

The time constants are

$$\tau_\pm = \frac{1}{1 \mp W_0}$$

Depending on the value of $W_0$, we get different dynamical regimes for $\alpha_+$, $\alpha_-$.

$|W_0| > 1 \Rightarrow$ Unstable exponential growth of $\alpha_+$ or $\alpha_-$

$|W_0| < 1 \Rightarrow$ Exponential approach

$W_0 = 1 \Rightarrow$ Perfect integrator of common mode

$W_0 = -1 \Rightarrow$ Perfect integrator of diffe~~n~~ence mode

$|W_0| \lesssim 1 \Rightarrow$ Leaky integrator in $\alpha_+$ or $\alpha_-$

We recover neural dynamics as

$$r = \begin{pmatrix} r_1 \\ r_2 \end{pmatrix} = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} \begin{pmatrix} \alpha_+ \\ \alpha_- \end{pmatrix} = \frac{1}{\sqrt{2}} \begin{pmatrix} \alpha_+ + \alpha_- \\ \alpha_+ - \alpha_- \end{pmatrix}$$

Both neurons participate in both modes.

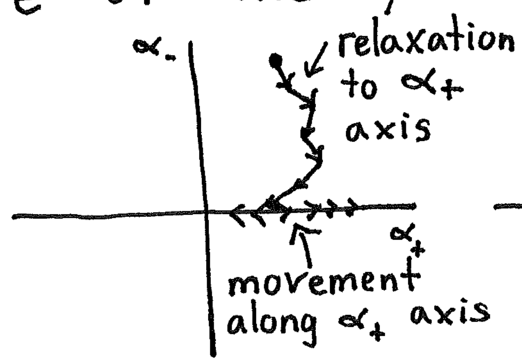
## Population activity patterns:

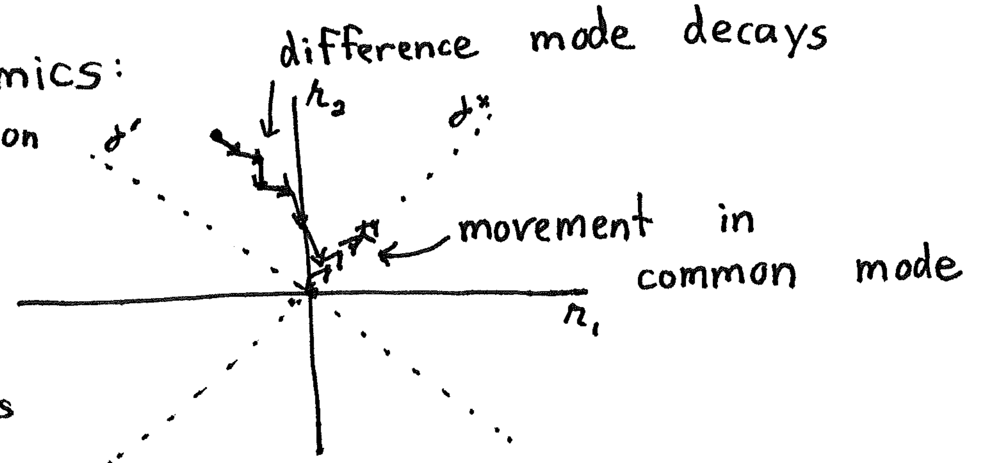
For concreteness, let's suppose $W_0 > 0$, $I_1 = I_2 = I_0$. Then

$$\frac{\tau}{1-W_0} \frac{d\alpha_+}{dt} = -\alpha_+ + \frac{\sqrt{2}\, I_0(t)}{1-W_0}$$

$$\frac{\tau}{1+W_0} \frac{d\alpha_-}{dt} = -\alpha_-$$

Nature of the dynamics:




A few critical points:

- The eigenvector components correspond to directions in activity space defined by V. Dynamics in $\alpha$ directly manifest as dynamics in the original activity space.

- $V$ and $V^{-1}$ correspond to basis transformation matrices. $(r_1, r_2)$ and $(\alpha_+, \alpha_-)$ are coordinates for the same underlying activity space. $r$-coordinates are especially meaningful for biological mechanisms; $\alpha$-coordinates for dynamics.
- The dynamics are approximately one-dimensional, because the difference mode decays to zero and stays there. Ongoing dynamics are confined to the common mode.

The first two points generally hold.

The third implies that we can reduce the dimensionality of the system

$$\begin{bmatrix} r_1(t) \\ r_2(t) \end{bmatrix} = V \begin{bmatrix} \alpha_1(t) \\ \alpha_2(t) \end{bmatrix} = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} \alpha_1(t) \\ \alpha_2(t) \end{bmatrix}$$

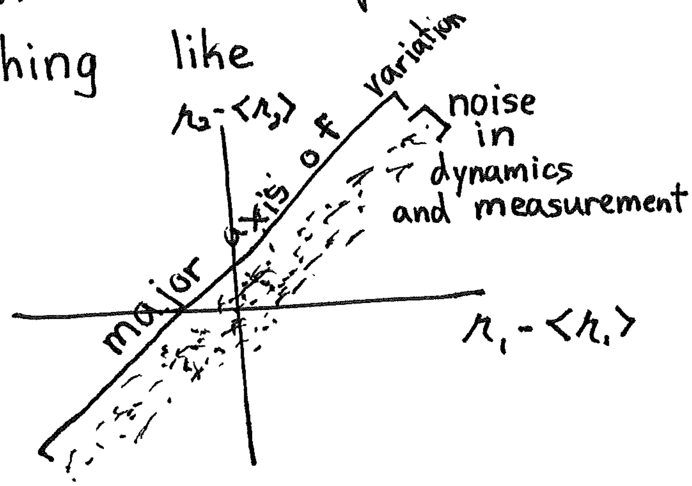
$$= \frac{1}{\sqrt{2}} \begin{bmatrix} \alpha_1(t) + \alpha_2(t) \\ \alpha_1(t) - \alpha_2(t) \end{bmatrix} = \underbrace{\frac{\alpha_1(t)}{\sqrt{2}} \begin{bmatrix} 1 \\ 1 \end{bmatrix}}_{\text{Very important}} + \underbrace{\frac{\alpha_2(t)}{\sqrt{2}} \begin{bmatrix} 1 \\ -1 \end{bmatrix}}_{\text{Negligible}}$$

$$\approx \underbrace{\frac{\alpha_1(t)}{\sqrt{2}} \begin{bmatrix} 1 \\ 1 \end{bmatrix}}_{\text{One-dimensional activity}}$$

This point applies to many other neural networks. However this is an impractical approach to dimensionality reduction, because we typically only know $r$, not $W$ and $I$.

# Principal component analysis (PCA):

Imagine you measure $r_1$ and $r_2$ at many time points. In our previous example, this would look something like




To reduce dimensionality from 2 to 1, need to find the direction of variation. So, we project $\delta r$ onto direction $v$ and compute variance of result

$$Var(v) = \langle (v^T \delta r)^2 \rangle = \langle v^T \delta r \, \delta r^T v \rangle$$

$$= v^T \underbrace{C_r}_{\text{Covariance matrix of activity}} v$$

Write covariance matrix in terms of its eigenvalues and eigenvectors

$$C_r = V D V^{-1} = V D V^T$$

where $V^{-1} = V^T$ because $C_r$ is symmetric. So

$$Var(v) = v^T V D V^T v = \sum_i \underbrace{\lambda_i}_{\text{eigenvalue}} \underbrace{(V^T v)_i^2}_{\text{Component along eigenvector}}$$

To maximize this, we align $v$ with the eigenvector of $C_r$ with largest eigenvalue. (set $\|v\| = 1$). We recover the reduced system.

More generally want to reduce $N$-dimensional system to $K < N$ dimensions. Second PC is vector orthogonal to first PC that maximizes variance. Since first PC is largest eigenvector of $C_r$, the previous equation becomes

$$Var(w) = \underbrace{\sum_{i=2}^{N}}_{\text{Skip PC1}} \lambda_i (V^T w)_i^2$$

Eigenvectors of $C_r$ are orthogonal

$$V^{-1} = V^T \Rightarrow \begin{bmatrix} \xi^{(1)T} \\ \vdots \\ \xi^{(N)T} \end{bmatrix} \begin{bmatrix} \xi^{(1)} & \xi^{(2)} & \cdots & \xi^{(N)} \end{bmatrix} = \begin{bmatrix} 1 & 0 & & 0 \\ 0 & 1 & & \\ & & \ddots & \\ 0 & & & 1 \end{bmatrix} \Rightarrow \xi^{(i)T} \xi^{(j)} = \delta_{ij}$$

So solution is to align PC2 to eigenvector of $C_r$ with second largest eigenvalue. And so on...

General PCA algorithm:

- Compute neural covariance matrix, $C_r$.
- Find its eigenvalues and eigenvectors.
- Reduce data to $K$ dimensions by projecting it onto the $K$ eigenvectors with largest eigenvalues.

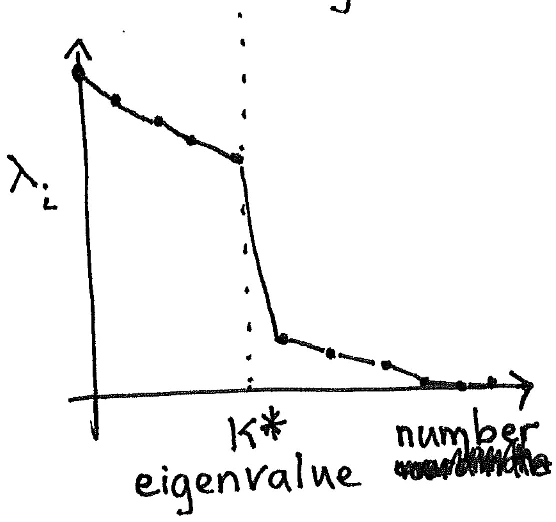
$$P_i = \xi^{(i)T} r$$

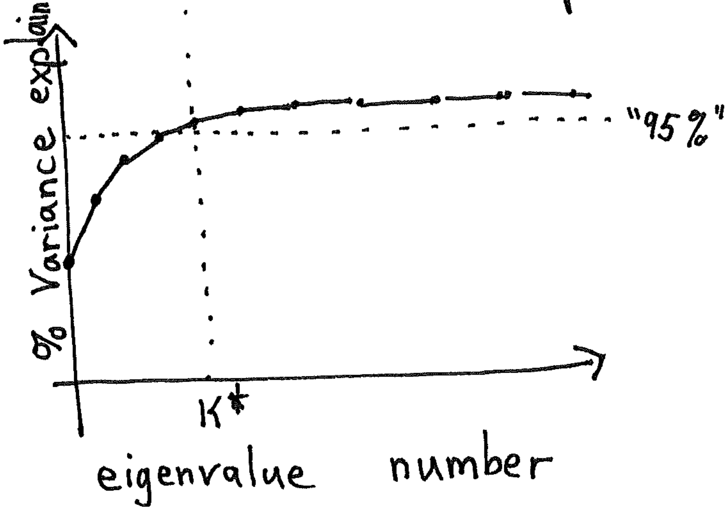
- Discard $N-K$ low-variance dimensions
- Can reconstruct "denoised" neural activity as

$$r \approx V \begin{pmatrix} P_1 \\ \vdots \\ P_K \\ 0 \end{pmatrix} = \sum_{i=1}^{K} P_i \, \xi^{(i)}$$

Determining the dimensionality:

"Cliff" in eigenvalues ; Cutoff in variance explained




% Variance explained

$$r_i^2 = \sum_{j=1}^{i} \lambda_i$$